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- Then raise the tube approximately about two times the first resonating length. Excite the tuning fork again and place it on the mouth of the tube.
- Change the height of the tube until the maximum sound is heard.
- Measure the length of air column at this position. This is taken as the second resonating length l_2
- We can now calculate the velocity of sound in air at room temperature by using the relation.

  $V = 2v(l_2 - l_1)$
- Repeat the experiment with forks of different frequency and calculate the velocity.
- The mean of the calculated values will give the velocity of sound in air at room temperature.

OBSERVATIONS

Sl. No.	Frequency of tuning fork v (Hz)	First resonating length $l_1(cm)$			Second resonating length $l_2(cm)$			$l_2 - l_1$ $(\times 10^{-2} m)$	Velocity of sound $V = 2v(l_2 - l_1)$ (m s^{-1})
		Trial 1	Trial 2	Mean	Trial 1	Trial 2	Mean		
1									
2									
3									
								Mean V =	

CALCULATION

Room temperature, t = _____ ° C

Velocity sound in air at room temperature, $V = 2v(l_2 - l_1)$ = _____ m s^{-1}

RESULT

Velocity of sound in air at room temperature, (V) = _____ m s^{-1}